Manual Calculation Worksheets

These are the worked solutions for the individual component (Section B of the assignment). The assignment requires the calculations to be handwritten and scanned, so each member copies their two files out by hand, following the steps with the source file open, and attaches the scans to Appendix A of the group report. Line numbers refer to the files in `SRE-A1/markdown/` at commit `0bf535bc`.

Member	File 1	File 2
Taha Yasir (i222429)	`markdown/extensions/def_list.py`	`markdown/extensions/meta.py`
Khizar (i222595)	`markdown/extensions/admonition.py`	`markdown/extensions/wikilinks.py`
Hadeed Abbas (i228780)	`markdown/postprocessors.py`	`markdown/extensions/legacy_attrs.py`

Rules to write at the top of your sheet

1. Comment density (SonarQube definition)

```
Comment density = comment lines / (comment lines + code lines) x 100

blank line     : contains only whitespace (not counted anywhere)
comment line   : a line that is only a # comment, or any non-blank line of a docstring
code line      : every other non-blank line (code + trailing # comment = code line)
```

2. Cyclomatic complexity (McCabe, per function)

```
CC = D + 1        D = number of decision points in the function

+1 for each:  if, elif, for, while, except, conditional expression (a if c else b),
              each `and` / `or`, each `for` and each `if` inside a comprehension
+0 for:       else, try, finally, return, break
A nested function is its own unit with its own +1.     File CC = sum over all functions.
```

3. Cognitive complexity (SonarSource, G. A. Campbell)

```
Start at 0 for each function.
Structural increment  +1 : if, elif, else, for, while, except, conditional expression,
                           each sequence of the same boolean operator (a and b and c = +1)
Nesting increment     +N : added to if, for, while, except, conditional expression.
                           N = number of enclosing if/elif/else, loop, except
                           or nested-function blocks.
                           elif, else and boolean sequences get NO nesting increment.
Increment written in the tables below = 1 + N.
```

The same rule set was run over all 397 functions of the package as a check: the cyclomatic rules reproduce radon's result for every function. The cognitive rules differ from the complexipy tool only where that tool adds points for comprehensions.

Taha Yasir (i222429) - File 1: markdown/extensions/def_list.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 28-29, 32, 35, 38, 40, 70, 88, 91-92, 95, 101, 104, 107-108, 111, 115, 120, 125-126	29
# comment	1-2, 4, 6-7, 9, 11, 13, 57-58, 62, 66, 72, 77, 79, 83, 96, 122	18
Docstring	15-16, 18-20, 31, 94, 98, 100, 103, 110, 113-114, 116-119, 121	18
Code	22, 24-27, 30, 33-34, 36-37, 39, 41-56, 59-61, 63-65, 67-69, 71, 73-76, 78, 80-82, 84-87, 89-90, 93, 97, 99, 102, 105-106, 109, 112, 123-124, 127-128	63
Total	**check: 29 + 18 + 18 + 63**	**128**

Line(s) 127: code followed by a trailing # comment, counted as code.

```
Comment lines   = 18 + 18 = 36
Code lines      = 63
Comment density = 36 / (36 + 63) x 100 = 36 / 99 x 100 = 36.4 %

Cross-check (radon raw): SLOC = 63 (matches the code count), blank = 29
```

Step 2. Cyclomatic complexity

DefListProcessor.run (lines 39-90)

Line	Decision point	Add	Running CC
-	function entry	1	1
43	comprehension 'for'	+1	2
44	comprehension 'if' x1	+1	3
47	if / elif	+1	4
51	if / elif	+1	5
56	if / elif	+1	6
56	boolean operator 'and' x1	+1	7
61	if / elif	+1	8
61	boolean operator 'and' x1	+1	9
71	if / elif	+1	10
71	boolean operator 'and' x1	+1	11
74	if / elif	+1	12
74	boolean operator 'and' x3	+3	15
80	for loop	+1	16
89	if / elif	+1	17
	CC = 16 decision points + 1		**17**

Tool cross-check: radon reports 17. Match.

Functions with no decision point (CC = 1 each): `test`, `create_item`, `extendMarkdown`, `makeExtension`. That is 4 x 1 = 4.

```
File cyclomatic complexity = 17 + 4 = 21
Highest function = 17      Average = 21 / 5 functions = 4.20
```

Step 3. Cognitive complexity (stepwise)

DefListProcessor.run (lines 39-90)

Line	Construct	Nesting level	Increment	Add	Running total
47	if	0	+1	+1	1
49	else	-	+1	+1	2
51	if	0	+1	+1	3
53	else	-	+1	+1	4
56	if	0	+1	+1	5
56	sequence of 'and'	-	+1	+1	6
61	if	0	+1	+1	7
61	sequence of 'and'	-	+1	+1	8
68	else	-	+1	+1	9
71	if	0	+1	+1	10
71	sequence of 'and'	-	+1	+1	11
74	if	1	+1 +1 nesting	+2	13
74	sequence of 'and'	-	+1	+1	14
76	else	-	+1	+1	15
80	for loop	0	+1	+1	16
89	if	0	+1	+1	17
	Cognitive complexity				**17**

Tool cross-check: complexipy reports 19. See the note below.

Functions with a straight-line body (cognitive complexity 0): `test`, `create_item`, `extendMarkdown`, `makeExtension`.

```
File cognitive complexity = 17        Highest function = 17
```

Note. Tool difference: complexipy reports 19 for `DefListProcessor.run`. It also counts the list comprehension on lines 43-44 (+1 for its `for`, +1 for its `if`). The SonarSource specification does not list comprehensions as incrementing structures, so the hand result is 17. State both numbers and this reason on your sheet.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
36.4 %	21 / 17	17 / 17

Taha Yasir (i222429) - File 2: markdown/extensions/meta.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 29, 31, 37-38, 41, 44, 48, 54, 57-58, 61, 90-91	22
# comment	1-2, 4, 6-7, 9, 11, 13, 32, 50, 83	11
Docstring	15-16, 18-20, 40, 43, 45-47, 49, 60, 63	13
Code	22, 24-28, 30, 33-36, 39, 42, 51-53, 55-56, 59, 62, 64-82, 84-89, 92-93	47
Total	**check: 22 + 11 + 13 + 47**	**93**

Line(s) 72, 87, 92: code followed by a trailing # comment, counted as code.

```
Comment lines   = 11 + 13 = 24
Code lines      = 47
Comment density = 24 / (24 + 47) x 100 = 24 / 71 x 100 = 33.8 %

Cross-check (radon raw): SLOC = 47 (matches the code count), blank = 22
```

Step 2. Cyclomatic complexity

MetaPreprocessor.run (lines 62-89)

Line	Decision point	Add	Running CC
-	function entry	1	1
66	if / elif	+1	2
66	boolean operator 'and' x1	+1	3
68	while loop	+1	4
71	if / elif	+1	5
71	boolean operator 'or' x1	+1	6
73	if / elif	+1	7
78	except clause	+1	8
82	if / elif	+1	9
82	boolean operator 'and' x1	+1	10
	CC = 9 decision points + 1		**10**

Tool cross-check: radon reports 10. Match.

Functions with no decision point (CC = 1 each): `extendMarkdown`, `reset`, `makeExtension`. That is 3 x 1 = 3.

```
File cyclomatic complexity = 10 + 3 = 13
Highest function = 10      Average = 13 / 4 functions = 3.25
```

Step 3. Cognitive complexity (stepwise)

MetaPreprocessor.run (lines 62-89)

Line	Construct	Nesting level	Increment	Add	Running total
66	if	0	+1	+1	1
66	sequence of 'and'	-	+1	+1	2
68	while loop	0	+1	+1	3
71	if	1	+1 +1 nesting	+2	5
71	sequence of 'or'	-	+1	+1	6

Line	Construct	Nesting level	Increment	Add	Running total
73	if	1	+1 +1 nesting	+2	8
78	except	2	+1 +2 nesting	+3	11
80	else	-	+1	+1	12
82	if	2	+1 +2 nesting	+3	15
82	sequence of 'and'	-	+1	+1	16
85	else	-	+1	+1	17
	Cognitive complexity				**17**

Tool cross-check: complexipy reports 17. Match.

Functions with a straight-line body (cognitive complexity 0): `extendMarkdown`, `reset`, `makeExtension`.

```
File cognitive complexity = 17        Highest function = 17
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
33.8 %	13 / 10	17 / 17

Khizar (i222595) - File 1: markdown/extensions/admonition.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 7, 10, 12, 14, 16-17, 21, 23, 27, 29, 35, 38-39, 42, 46, 50, 54, 56-57, 59, 64, 67, 69, 72, 75, 78, 80, 83, 91, 93, 106, 111, 119, 122, 128, 130, 132, 137, 141, 149, 165, 167, 169, 175, 189-190	48
# comment	1-2, 4, 6, 8-9, 11, 13, 15, 52, 84, 97-98, 107-108, 112-114, 159, 171-173, 180-182, 185-186	27
Docstring	18-20, 22, 24-26, 41, 44-45, 47-49, 51, 66, 74, 76-77, 79	19
Code	28, 30-34, 36-37, 40, 43, 53, 55, 58, 60-63, 65, 68, 70-71, 73, 81-82, 85-90, 92, 94-96, 99-105, 109-110, 115-118, 120-121, 123-127, 129, 131, 133-136, 138-140, 142-148, 150-158, 160-164, 166, 168, 170, 174, 176-179, 183-184, 187-188, 191-192	98
Total	**check: 48 + 27 + 19 + 98**	**192**

Line(s) 36, 145, 191: code followed by a trailing # comment, counted as code.

```
Comment lines   = 27 + 19 = 46
Code lines      = 98
Comment density = 46 / (46 + 98) x 100 = 46 / 144 x 100 = 31.9 %

Cross-check (radon raw): SLOC = 98 (matches the code count), blank = 48
```

Step 2. Cyclomatic complexity

AdmonitionProcessor.parse_content (lines 73-129)

Line	Decision point	Add	Running CC
-	function entry	1	1
85	if / elif	+1	2
94	if / elif	+1	3
94	boolean operator 'or' x2	+2	5
101	while loop	+1	6
102	if / elif	+1	7
103	boolean operator 'and' x3	+3	10
110	conditional expression (x if c else y)	+1	11
120	if / elif	+1	12
123	if / elif	+1	13
	CC = 12 decision points + 1		**13**

Tool cross-check: radon reports 13. Match.

AdmonitionProcessor.test (lines 131-136)

Line	Decision point	Add	Running CC
-	function entry	1	1
133	if / elif	+1	2
	CC = 1 decision points + 1		**2**

Tool cross-check: radon reports 2. Match.

AdmonitionProcessor.run (lines 138-174)

Line	Decision point	Add	Running CC
-	function entry	1	1
142	if / elif	+1	2

Line	Decision point	Add	Running CC
143	if / elif	+1	3
150	if / elif	+1	4
154	if / elif	+1	5
160	if / elif	+1	6
160	boolean operator 'and' x1	+1	7
170	if / elif	+1	8
	CC = 7 decision points + 1		**8**

Tool cross-check: radon reports 8. Match.

AdmonitionProcessor.get_class_and_title (lines 176-188)

Line	Decision point	Add	Running CC
-	function entry	1	1
179	if / elif	+1	2
184	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

Functions with no decision point (CC = 1 each): `extendMarkdown`, `__init__`, `makeExtension`. That is 3 x 1 = 3.

```
File cyclomatic complexity = 13 + 2 + 8 + 3 + 3 = 29
Highest function = 13      Average = 29 / 7 functions = 4.14
```

Step 3. Cognitive complexity (stepwise)

AdmonitionProcessor.parse_content (lines 73-129)

Line	Construct	Nesting level	Increment	Add	Running total
85	if	0	+1	+1	1
94	if	0	+1	+1	2
94	sequence of 'or'	-	+1	+1	3
96	else	-	+1	+1	4
101	while loop	1	+1 +1 nesting	+2	6
102	if	2	+1 +2 nesting	+3	9
103	sequence of 'and'	-	+1	+1	10
110	conditional expression	3	+1 +3 nesting	+4	14
117	else	-	+1	+1	15
120	if	1	+1 +1 nesting	+2	17
123	if	1	+1 +1 nesting	+2	19
	Cognitive complexity				**19**

Tool cross-check: complexipy reports 19. Match.

AdmonitionProcessor.test (lines 131-136)

Line	Construct	Nesting level	Increment	Add	Running total
133	if	0	+1	+1	1
135	else	-	+1	+1	2
	Cognitive complexity				**2**

Tool cross-check: complexipy reports 2. Match.

AdmonitionProcessor.run (lines 138-174)

Line	Construct	Nesting level	Increment	Add	Running total
142	if	0	+1	+1	1
143	if	1	+1 +1 nesting	+2	3
147	else	-	+1	+1	4
150	if	0	+1	+1	5
154	if	1	+1 +1 nesting	+2	7
158	else	-	+1	+1	8
160	if	1	+1 +1 nesting	+2	10
160	sequence of 'and'	-	+1	+1	11
170	if	0	+1	+1	12
	Cognitive complexity				**12**

Tool cross-check: complexipy reports 12. Match.

AdmonitionProcessor.get_class_and_title (lines 176-188)

Line	Construct	Nesting level	Increment	Add	Running total
179	if	0	+1	+1	1
184	elif	-	+1	+1	2
	Cognitive complexity				**2**

Tool cross-check: complexipy reports 2. Match.

Functions with a straight-line body (cognitive complexity 0): `extendMarkdown`, `__init__`, `makeExtension`.

```
File cognitive complexity = 19 + 2 + 12 + 2 = 35        Highest function = 19
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
31.9 %	29 / 13	35 / 19

Khizar (i222595) - File 2: markdown/extensions/wikilinks.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 29-30, 35-36, 39, 49, 52, 56, 60, 66-67, 70, 74, 88, 102-103	25
# comment	1-2, 4, 6-7, 9, 11, 13, 58, 61	10
Docstring	15-16, 18-20, 32, 38, 47, 51, 53-55, 57, 69, 90	15
Code	22, 24-28, 31, 33-34, 37, 40-46, 48, 50, 59, 62-65, 68, 71-73, 75-87, 89, 91-101, 104-105	55
Total	**check: 25 + 10 + 15 + 55**	**105**

Line(s) 104: code followed by a trailing # comment, counted as code.

```
Comment lines   = 10 + 15 = 25
Code lines      = 55
Comment density = 25 / (25 + 55) x 100 = 25 / 80 x 100 = 31.2 %

Cross-check (radon raw): SLOC = 55 (matches the code count), blank = 25
```

Step 2. Cyclomatic complexity

WikiLinksInlineProcessor.handleMatch (lines 75-87)

Line	Decision point	Add	Running CC
-	function entry	1	1
76	if / elif	+1	2
83	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

WikiLinksInlineProcessor._getMeta (lines 89-101)

Line	Decision point	Add	Running CC
-	function entry	1	1
94	if / elif	+1	2
95	if / elif	+1	3
97	if / elif	+1	4
99	if / elif	+1	5
	CC = 4 decision points + 1		**5**

Tool cross-check: radon reports 5. Match.

Functions with no decision point (CC = 1 each): `build_url`, `__init__`, `extendMarkdown`, `__init__`, `makeExtension`. That is 5 x 1 = 5.

```
File cyclomatic complexity = 3 + 5 + 5 = 13
Highest function = 5      Average = 13 / 7 functions = 1.86
```

Step 3. Cognitive complexity (stepwise)

WikiLinksInlineProcessor.handleMatch (lines 75-87)

Line	Construct	Nesting level	Increment	Add	Running total
76	if	0	+1	+1	1

Line	Construct	Nesting level	Increment	Add	Running total
83	if	1	+1 +1 nesting	+2	3
85	else	-	+1	+1	4
	Cognitive complexity				**4**

Tool cross-check: complexipy reports 4. Match.

WikiLinksInlineProcessor._getMeta (lines 89-101)

Line	Construct	Nesting level	Increment	Add	Running total
94	if	0	+1	+1	1
95	if	1	+1 +1 nesting	+2	3
97	if	1	+1 +1 nesting	+2	5
99	if	1	+1 +1 nesting	+2	7
	Cognitive complexity				**7**

Tool cross-check: complexipy reports 7. Match.

Functions with a straight-line body (cognitive complexity 0): `build_url`, `__init__`, `extendMarkdown`, `__init__`, `makeExtension`.

```
File cognitive complexity = 4 + 7 = 11        Highest function = 7
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
31.2 %	13 / 5	11 / 7

Hadeed Abbas (i228780) - File 1: markdown/postprocessors.py

Step 1. Comment density

Category	Line numbers	Count
Blank	2, 4, 8, 13, 17, 19, 21, 26, 28, 30, 34, 37-38, 45-46, 50, 53, 55, 57, 63, 66-67, 70, 72, 87, 94, 104, 108-109, 112, 116-117, 124, 126, 129	35
# comment	1, 3, 5-7, 9-12, 14-16, 18, 100	14
Docstring	20, 22-25, 27, 40, 48-49, 51-52, 54, 56, 59-62, 64, 69, 74, 96, 106, 111, 123	24
Code	29, 31-33, 35-36, 39, 41-44, 47, 58, 65, 68, 71, 73, 75-86, 88-93, 95, 97-99, 101-103, 105, 107, 110, 113-115, 118-122, 125, 127-128, 130-131	58
Total	**check: 35 + 14 + 24 + 58**	**131**

Line(s) 35, 65: code followed by a trailing # comment, counted as code.

```
Comment lines   = 14 + 24 = 38
Code lines      = 58
Comment density = 38 / (38 + 58) x 100 = 38 / 96 x 100 = 39.6 %

Cross-check (radon raw): SLOC = 58 (matches the code count), blank = 35
```

Step 2. Cyclomatic complexity

RawHtmlPostprocessor.run (lines 73-93)

Line	Decision point	Add	Running CC
-	function entry	1	1
88	if / elif	+1	2
	CC = 1 decision points + 1		**2**

Tool cross-check: radon reports 2. Match.

RawHtmlPostprocessor.run > nested function substitute_match (lines 75-86)

Line	Decision point	Add	Running CC
-	function entry	1	1
76	if / elif	+1	2
81	if / elif	+1	3
84	if / elif	+1	4
84	boolean operator 'or' x1	+1	5
	CC = 4 decision points + 1		**5**

Tool cross-check: radon reports 5. Match.

RawHtmlPostprocessor.isblocklevel (lines 95-103)

Line	Decision point	Add	Running CC
-	function entry	1	1
98	if / elif	+1	2
99	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

Functions with no decision point (CC = 1 each): `build_postprocessors`, `run`, `stash_to_string`, `run`, `unescape`, `run`. That is 6 x 1 = 6.

```
File cyclomatic complexity = 2 + 5 + 3 + 6 = 16
Highest function = 5      Average = 16 / 9 functions = 1.78
```

Step 3. Cognitive complexity (stepwise)

RawHtmlPostprocessor.run (lines 73-93)

Line	Construct	Nesting level	Increment	Add	Running total
76	if	1	+1 +1 nesting	+2	2
78	else	-	+1	+1	3
81	if	1	+1 +1 nesting	+2	5
84	if	1	+1 +1 nesting	+2	7
84	sequence of 'or'	-	+1	+1	8
88	if	0	+1	+1	9
92	else	-	+1	+1	10
	Cognitive complexity				**10**

Tool cross-check: complexipy reports 10. Match.

RawHtmlPostprocessor.isblocklevel (lines 95-103)

Line	Construct	Nesting level	Increment	Add	Running total
98	if	0	+1	+1	1
99	if	1	+1 +1 nesting	+2	3
	Cognitive complexity				**3**

Tool cross-check: complexipy reports 3. Match.

Functions with a straight-line body (cognitive complexity 0): `build_postprocessors`, `run`, `stash_to_string`, `run`, `unescape`, `run`.

```
File cognitive complexity = 10 + 3 = 13        Highest function = 10
```

Note. The nested function `substitute_match` (lines 75-86) is a separate unit for cyclomatic complexity (it has its own +1), but for cognitive complexity it is charged to `run`, with everything inside it one nesting level deeper. That is why line 76 scores 2, not 1.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
39.6 %	16 / 5	13 / 10

Hadeed Abbas (i228780) - File 2: markdown/extensions/legacy_attrs.py

Step 1. Comment density

Category	Line numbers	Count
Blank	2, 4, 8, 13, 17, 19, 22, 30, 32, 37, 40-41, 43-44, 56, 62-63, 67, 71, 75-76	21
# comment	1, 3, 5-7, 9-12, 14-16, 18, 73	14
Docstring	20-21, 23-29, 47, 58, 66, 68-70, 72	16
Code	31, 33-36, 38-39, 42, 45-46, 48-55, 57, 59-61, 64-65, 74, 77-78	27
Total	**check: 21 + 14 + 16 + 27**	**78**

Line(s) 38, 42, 77: code followed by a trailing # comment, counted as code.

```
Comment lines   = 14 + 16 = 30
Code lines      = 27
Comment density = 30 / (30 + 27) x 100 = 30 / 57 x 100 = 52.6 %

Cross-check (radon raw): SLOC = 27 (matches the code count), blank = 21
```

Step 2. Cyclomatic complexity

LegacyAttrs.run (lines 46-55)

Line	Decision point	Add	Running CC
-	function entry	1	1
48	for loop	+1	2
50	if / elif	+1	3
52	if / elif	+1	4
52	boolean operator 'and' x1	+1	5
54	if / elif	+1	6
54	boolean operator 'and' x1	+1	7
	CC = 6 decision points + 1		**7**

Tool cross-check: radon reports 7. Match.

Functions with no decision point (CC = 1 each): `handleAttributes`, `attributeCallback`, `extendMarkdown`, `makeExtension`. That is 4 x 1 = 4.

```
File cyclomatic complexity = 7 + 4 = 11
Highest function = 7      Average = 11 / 5 functions = 2.20
```

Step 3. Cognitive complexity (stepwise)

LegacyAttrs.run (lines 46-55)

Line	Construct	Nesting level	Increment	Add	Running total
48	for loop	0	+1	+1	1
50	if	1	+1 +1 nesting	+2	3
52	if	1	+1 +1 nesting	+2	5
52	sequence of 'and'	-	+1	+1	6
54	if	1	+1 +1 nesting	+2	8
54	sequence of 'and'	-	+1	+1	9
	Cognitive complexity				**9**

Tool cross-check: complexipy reports 9. Match.

Functions with a straight-line body (cognitive complexity 0): `handleAttributes`, `extendMarkdown`, `makeExtension`.

```
File cognitive complexity = 9        Highest function = 9
```

Note. The nested function `attributeCallback` (line 59) has no decision point: cyclomatic 1, and it adds nothing to the cognitive complexity of `handleAttributes`.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
52.6 %	11 / 7	9 / 9